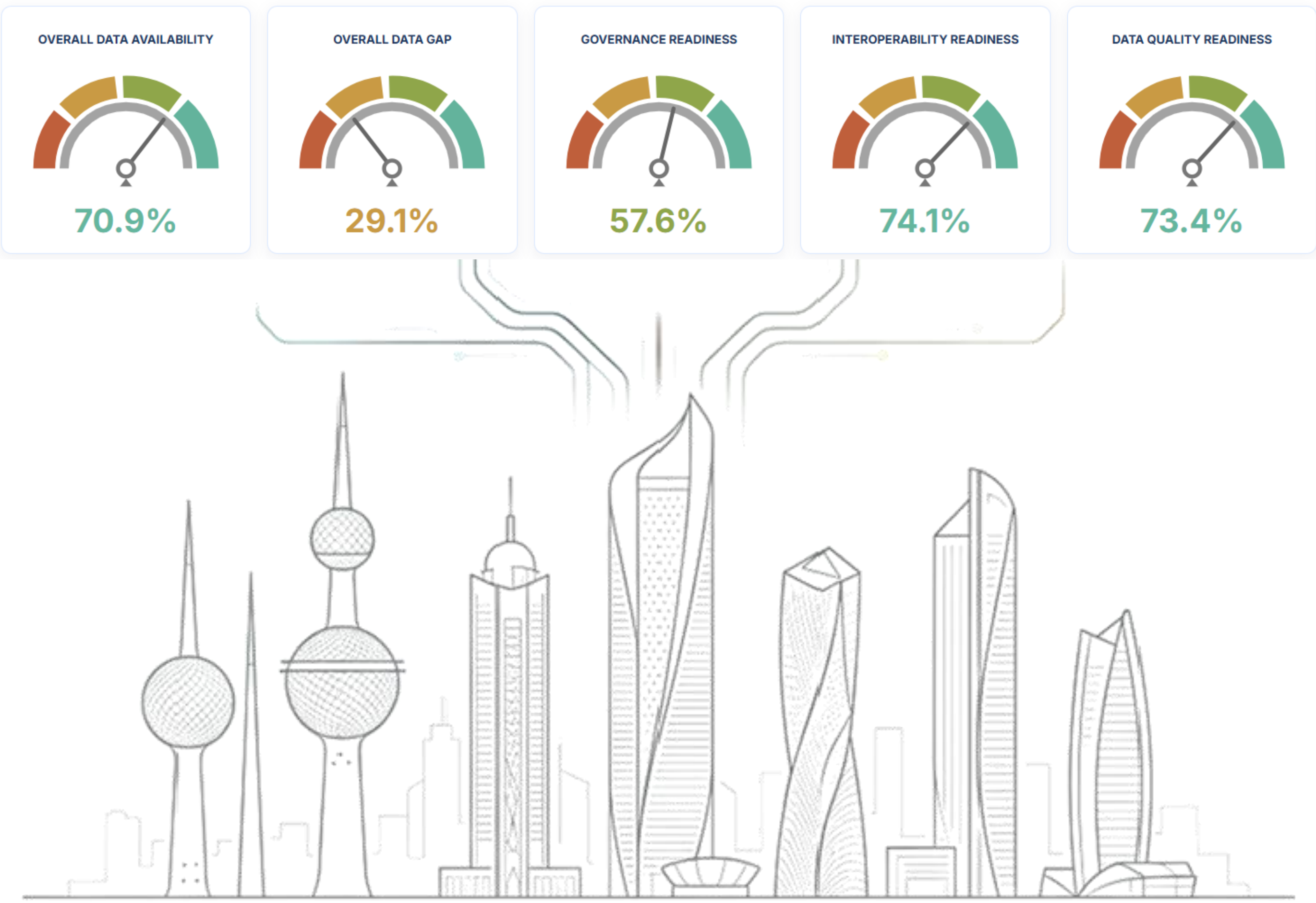
Situational Analysis of Environmental Data Stakeholders

Evidence-based | Governance-driven | Implementation-oriented



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1 Executive Summary

The data received from stakeholder questionnaire responses shows that environmental data are already being produced, used and exchanged across various institutional landscape. The challenge is to turn dispersed institutional data management practices to a more coherent, validated and nationally governed environmental data management system.

The respondent profile confirms that the NEDMP must be developed as a national coordination framework. Government institutions account for 71.9 per cent of respondents, while the private sector, oil sector, statistical entities and regional or international organizations also form part of the data landscape. Institutions are performing multiple roles at once: 59.4 per cent identify as data owners, 56.3 per cent as data producers, 46.9 per cent as data storage or repository actors, 46.9 per cent as users, and 40.6 per cent as coordinators or hubs for data exchange. This is a diverse institutional fabric. Nevertheless, without clear ownership, stewardship and decision rights, it can also become more challenging.

The overall data received through the assessment questionnaire presents a cautiously positive picture. Overall data availability stands at 70.9 per cent, making a data gap of 29.1 per cent. Interoperability readiness is relatively strong at 74.1 per cent, while data quality readiness reaches 73.4 per cent. Governance readiness, however, is lower at 57.6 per cent. This gap suggests that Kuwait has a functioning environmental data base and some technical readiness, but the rules, responsibilities and institutional arrangements needed to make the system work as one national platform need more enforcement.

Data production is one of the strongest foundations for the NEDMP. Around 81.5 per cent of institutions report that they produce environmental data periodically or regularly. Field monitoring data is the most common source, reported by 86.7 per cent of respondents, followed by measuring station data at 63.3 per cent and survey or statistical data at 60.0 per cent. GIS data, modelling and forecasting data, and satellite imagery are also present, though at lower levels. This diversity is valuable. It also means that the NEDMP must be capable of managing observational, statistical, spatial, modelling and remote-sensing datasets under one harmonized framework.

The report shows that data completeness, update frequency and historical time-series coverage vary across institutions. Some datasets are strong and mature; others are partial, irregular or not yet suitable for national reporting and trend analysis. The NEDMP should therefore establish a national dataset inventory, define minimum metadata requirements, and set clear QA/QC rules before data are integrated, published or reused for policy decisions.

Data sharing presents a more complex picture. Formal data accessibility policies are reported by 78.1 per cent of institutions, which indicates that a policy foundation already exists. However, open-data practice remains limited: 40.6 per cent of institutions report no open-data practices, and 25.0 per cent report only limited open data. The main barriers are data sensitivity, reported by 82.8 per cent of respondents, and legal restrictions, reported by 65.5 per cent. This calls for nuance. Kuwait does not need an open-data approach that ignores sensitivity; it needs a governed access model that distinguishes between public data, internal data, restricted data and confidential data.

Data exchange is active, but still procedural. Official requests remain the dominant mechanism, used by 81.3 per cent of respondents. Electronic portals are used by 56.3 per cent, periodic report publication by 46.9 per cent, electronic linking systems by 43.8 per cent, memoranda of understanding by 37.5 per cent, and bilateral exchange agreements by 31.3 per cent. Only 40.0 per cent of institutions report standardized machine-readable formats, while 33.3 per cent report partial standardization. This shows that interoperability is emerging, but not yet settled. The NEDMP should gradually shift the system from request-based exchange to structured, recurrent and machine-readable data flows.

Quality and documentation are important element for the credibility of the whole system. QA/QC procedures are fully implemented by 61.3 per cent of respondents, partially implemented by 19.4 per cent, and absent in 16.1 per cent. Reliability is also mixed, with 38.7 per cent rating their data as high reliability and another 38.7 per cent as good. Data that contains no metadata, method documentation or audit trails should not be treated as ready for national reporting, public dissemination or advanced analytics.

Systems and infrastructure also vary significantly. Institutions use different environmental data management systems, and their infrastructure efficiency is uneven. Formal backup and disaster recovery plans tested regularly are reported by 53.1 per cent of respondents, while 31.3 per cent report only partial procedures and 12.5 per cent report no formal backup strategy. Before Kuwaiti institutions move toward full platform integration, the NEDMP should map existing systems, assess digital readiness, and define minimum national requirements for backup, cybersecurity, hosting, disaster recovery and data continuity.

Human capacity is another critical isue. Some institutions have a strong specialist base, with 41.9 per cent reporting more than 10 specialists working on environmental data and indicators. At the same time, 38.7 per cent report fewer than five specialists, and staff sufficiency is only partial in 41.9 per cent of responses. Training needs cut across data governance, metadata, QA/QC, GIS, databases, reporting, APIs and digital platforms. Capacity development should therefore not be treated as a side activity. It is a core implementation pillar.

Governance and security findings reinforce the same message. Information management and data governance policies exist in 56.3 per cent of institutions and partially exist in 21.9 per cent, while 12.5 per cent report no such policies. Data security systems and breach response arrangements are reported by 54.8 per cent of institutions, with 24.1 per cent reporting partial arrangements. Standard Operating Procedures are in place in only 35.5 per cent of institutions, partially in 32.3 per cent, and absent in another 32.3 per cent. The NEDMP should therefore move quickly to establish national SOP templates, security protocols, escalation procedures and compliance review mechanisms.

Environmental reporting is already a major driver of data demand, with institutions participating in national reports such as Kuwait State of the Environment Report, international reporting processes (Global Environment Outlook (GEO) and West Asia Outlook Report) and sectoral reporting obligations. Mandatory reporting data are fully available in 56.3 per cent of responses and partially available in 37.5 per cent. Environmental data are also used in the policy cycle: 67.7 per cent of institutions use data for monitoring and evaluation, while 58.1 per cent use it in early policy design and another 58.1 per cent in policy implementation. This confirms that environmental data are not merely stored; they are already shaping decisions. The NEDMP can make that influence more consistent, transparent and defensible.

This report therefore points to a clear implementation pathway. First, to build the governance floor: roles, rules, dataset registry, metadata, QA/QC, access classification and institutional focal points. Second, to standardize priority datasets and reporting flows, especially those linked to national environmental reporting and policy needs. Third, to strengthen infrastructure readiness and platform integration, including the role of eMISK and future national environmental data platforms. Finally, when the foundations are stable, move toward automated exchange, dashboards, advanced analytics and responsible AI-enabled applications.

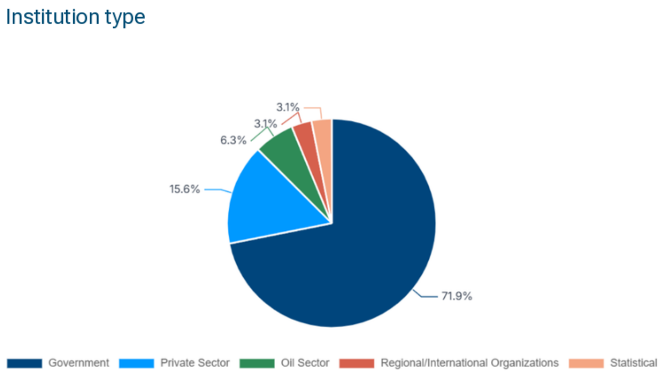
The central conclusion is that Kuwait has the institutional participation, data production base and strategic demand required to implement the NEDMP. What is needed is organized sequencing. Governance first. Standards next. Digital integration after that. With this approach, the NEDMP can become a practical instrument for stronger environmental governance, better reporting, more transparent access, and more informed decision-making for Kuwait’s environmental future.

2 Introduction

The submitted environmental data assessment questionnaire responses from various relevant stakeholders indicate that Kuwait has a broad and active environmental data systems. Government institutions represent the largest responding group, while private sector, oil sector, statistical and regional/international actors also contribute to the national data landscape.

The responses (as received from institutions) show existing capacity in environmental data production. A large part of institutions produces environmental data periodically or regularly, and most depend on formal monitoring data, measurement stations and survey or statistical data. This creates a strong foundation for a national database system, common metadata standards and structured reporting workflows.

The important challenge is the inconsistency of level maturity of governance, interoperability, accessibility, open data, metadata, QA/QC, institutional coordination, digital integration, human capacity and sustainable financing. The NEDMP should therefore prioritize governance rules, standard operating procedures, stewardship responsibilities and phased digital integration before moving to more unified digitized system, APIs and artificial intelligence applications.



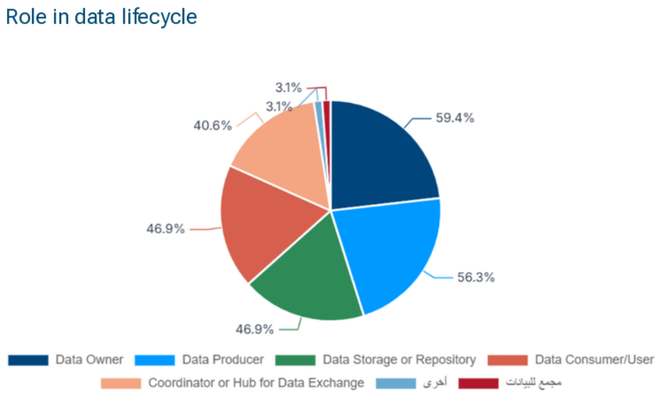


Figure 1 Institutional respondent profile and roles in the national environmental data lifecycle.

Table 1 Policy-maker summary of questionnaire evidence and NEDMP implications Source

Evidence area	Main signal from responses	NEDMP implication
Institutional coverage	Government entities dominate the respondent profile, with participation from other institutional types.	Adopt a national inter-institutional governance model.
Data production	Regular environmental data production and strong monitoring data sources are visible.	Build a national dataset registry and reporting workflows around existing data assets.
Data sharing	Public availability, open data and formal access rules remain inconsistent.	Develop data classification, publication and controlled release protocols.
Interoperability	Official requests and portals are used, but APIs and standardized exchange are not universal.	Phase in common formats, metadata templates and system integration standards.
Capacity and sustainability	Training, financing and staffing gaps appear across responses.	Create a funded capacity-building and sustainability programme.

3 Institution Identification

3.1 Institutions Types and Overall Analysis

The respondent profile is led by government institutions (71.9 per cent), followed by private sector entities (15.6 per cent), oil sector entities (6.3 per cent), and smaller shares from regional/international organizations and statistical entities.

Results received from stakeholders show that the institutional landscape is multi-actor and multi-role. This confirms that the NEDMP cannot be implemented as a single-agency technical project; it requires national data governance, stewardship and coordination arrangements.

Table 2 Respondent institutional type

INSTITUTION TYPE	RESPONSES	PERCENTAG
Government	23	71.9%
Regional/International Organizations	1	3.1%
Private Sector	5	15.6%
Oil Sector	2	6.3%
Statistical	1	3.1%

The integration of all the responses indicates that Kuwaiti institutions show a generally positive baseline for the NEDMP, Respondents perform multiple roles in the environmental data lifecycle: 59.4 per cent identify as data owners, 56.3 per cent as data producers, 46.9 per cent as data storage or repository actors, 46.9 per cent as data consumers/users, and 40.6 per cent as coordinators or hubs for data exchange.

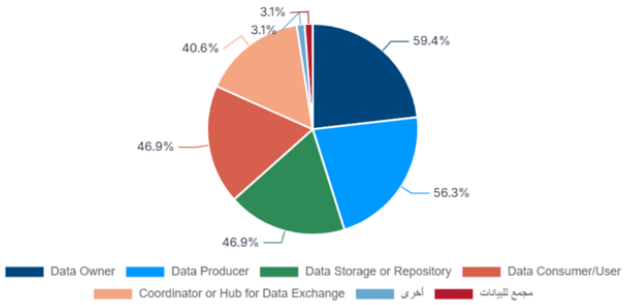


Figure 2 Institutional Role within environmental data lifecycle

with overall data availability reaching 70.9% and a remaining data gap of 29.1%. Interoperability readiness is relatively strong at 74.1%, closely followed by data quality readiness at 73.4%, suggesting that many institutions already have partial capacity to support data exchange and quality improvement. Governance readiness is lower at 57.6%, indicating that the main priority for NEDMP implementation should be strengthening institutional roles, data ownership, coordination mechanisms, standards, and formal governance procedures.

Accordingly, Governance takes preference in implementation over technology. When roles, standards, and procedures are well-defined, then the existing preparedness in environmental data quality and interoperability can be transformed into a functional national environmental data management system.

Figure 3 Overall Environmental Data Management Dashboard

Table 2. Institution Identification - key implications for NEDMP implementation

Results	Impact on NEDMP	Implementation priority
Government institutions represent 71.9% of respondents.	NEDMP must operate through public-sector coordination and formal mandates.	Confirm government focal points and official data custodians.
Respondents play multiple data lifecycle roles.	One institution may be producer, owner, user and repository.	Use role-based governance instead of institution-only classification.
Coordinator/hub role appears less common than ownership/production.	Interoperability may be limited by weak coordination hubs.	Create dedicated data coordination mechanisms under NEDMP.

3.2 Environmental Data Production

Responses show a strong national production base: 81.5 per cent of institutions reported producing environmental data periodically or regularly.

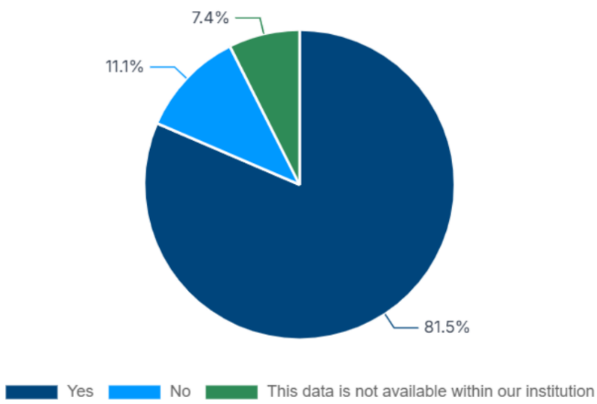


Figure 4 data production

However, data sources varied, led by field monitoring data (86.7 per cent), measuring station data (63.3 per cent), survey/statistical data (60.0 per cent), GIS data (36.7 per cent), modelling/forecasting data (33.3 per cent) and satellite imagery (26.7 per cent). Other minor sources were excluded.

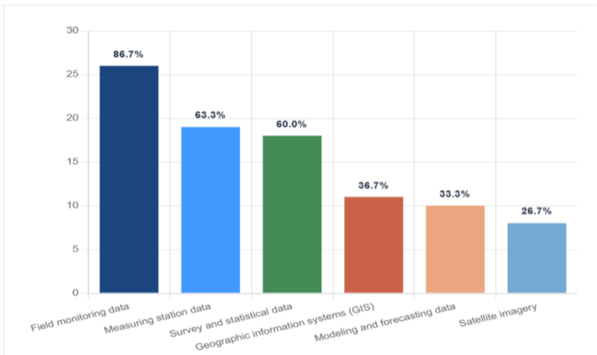


Figure 6 Sources of environmental data

Data update frequency is mixed, with monthly and irregular updates are the most visible categories. Data completeness is generally positive, with 40.0 per cent excellent and 30.0 per cent good; however, fair completeness and partial time-series practices indicate the need for harmonized standards.

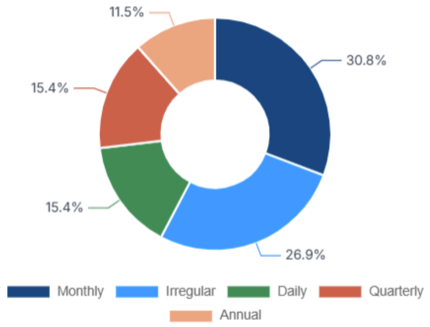


Figure 5 Frequency of environmental data updates

The national system has enough production capacity to support the NEDMP, but the diversity of sources, update cycles and completeness levels requires data classification, metadata, quality thresholds and recurring reporting schedules.

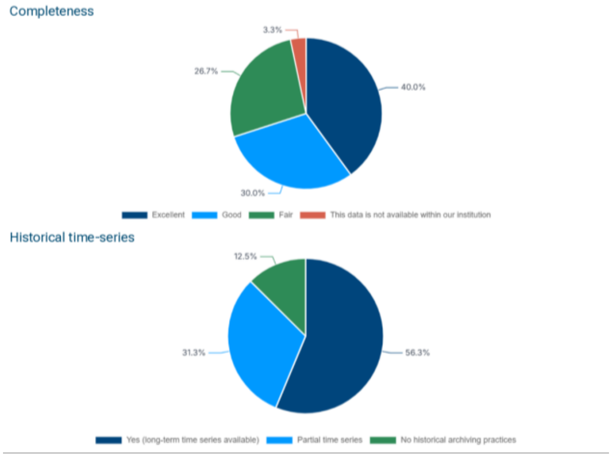


Figure 7 completeness and historical time-series availability.

Implementation Recommendation

Develop a national dataset inventory and indicator catalogue that records producer, source type, update frequency, completeness, time-series status, and responsible custodian for each priority dataset.

Table 3. Environmental Data Production - key implications for NEDMP implementation Source: Uploaded all-institutions questionnaire response analysis.

Results from questionnaire	Impact on NEDMP	Implementation priority
81.5% of institutions produce data periodically or regularly.	The NEDMP can build on an active data production base.	Prioritize inventory and classification of existing datasets.
Monitoring, station and survey/statistical data are major sources.	The plan should cover observational, statistical and spatial data together.	Adopt common metadata and QA/QC rules across source types.
Completeness and time-series coverage are uneven.	Trend analysis and reporting may be constrained.	Define minimum completeness and continuity requirements.

4 Data Sharing

The responses indicate that environmental data sharing exists but is not yet fully systematic. Public availability is split between upon-request access, partial availability, full availability and non-availability. Formal data accessibility policies are reported by 78.1 per cent of institutions, but open-data practice is weaker: 40.6 per cent report no open-data practices and 25.0 per cent report limited open data.

The main constraints are data sensitivity (82.8 per cent) and legal restrictions (65.5 per cent), while technical infrastructure, human resources and institutional conditions also appear as constraints.

The NEDMP should distinguish between data that can be fully open, data released on request, restricted intergovernmental data, and confidential/sensitive data. Without classification rules, public access and transparency will remain inconsistent.

NEDMP implementation recommendation

Develop public/internal/restricted/confidential data rules, a controlled-release protocol, publication approval workflow and legal review mechanism for public environmental data dissemination.

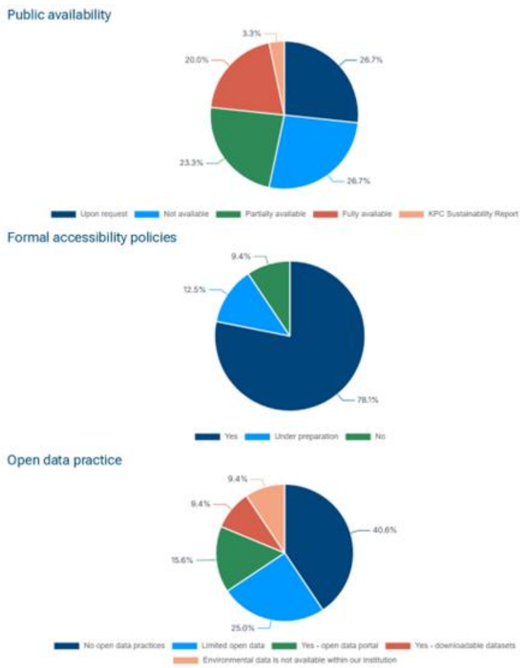


Figure 8 Public availability, data accessibility policy status and open-data practices

Table 3 Data Sharing - key implications for NEDMP implementation

Results from questionnaire	Impact on NEDMP	Implementation priority
78.1% report formal accessibility policies.	The policy foundation exists but needs harmonization.	Prepare a national data access policy and common access request procedure.
40.6% report no open-data practices.	Transparency and reuse are limited.	Create open-data readiness criteria and publishable dataset categories.
Data sensitivity and legal restrictions dominate constraints.	Access must be governed, not ad hoc.	Use classification and legal clearance before publication.

5 Data Exchange

Current data exchange is dominated by official requests (81.3 per cent), electronic portals (56.3 per cent), periodic report publication (46.9 per cent), electronic linking systems (43.8 per cent), memoranda of understanding (37.5 per cent), bilateral exchange agreements (31.3 per cent), shared cloud storage (18.8 per cent) and database publication (15.6 per cent).

Although 40.0 per cent report standardized machine-readable formats, 33.3 per cent indicate partial standardization and several responses point to manual or institution-specific arrangements. Coordination and external system integration capacity remain mixed.

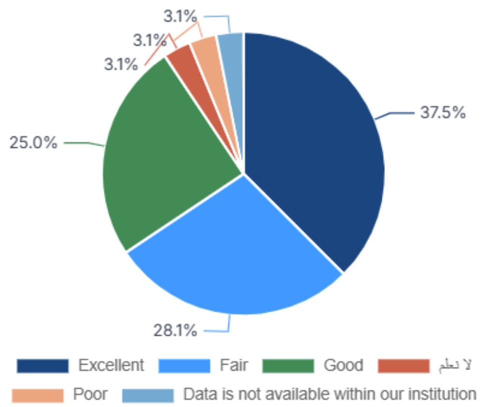


Figure 10 Assessment of the level of institutional coordination of data with other institutions

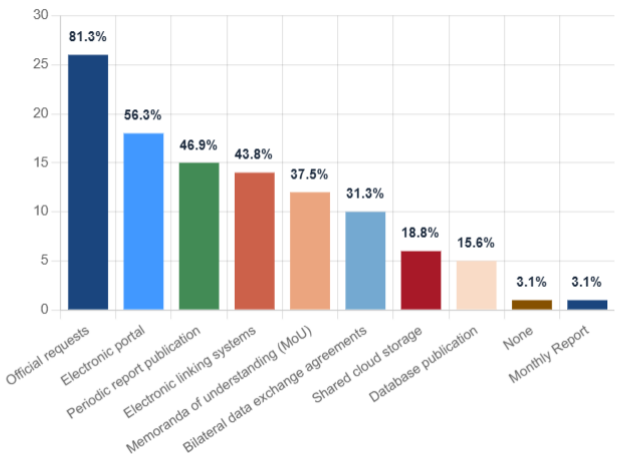


Figure 9 Current data exchange mechanisms

The exchange ecosystem is active but fragmented. The NEDMP should shift the system from request-based exchange towards governed, machine-readable, recurrent and interoperable data flows.

NEDMP implementation recommendation

Adopt national data exchange protocols, standard formats, API-readiness criteria, MoU templates, data-sharing agreements and a phased interoperability roadmap for priority datasets.

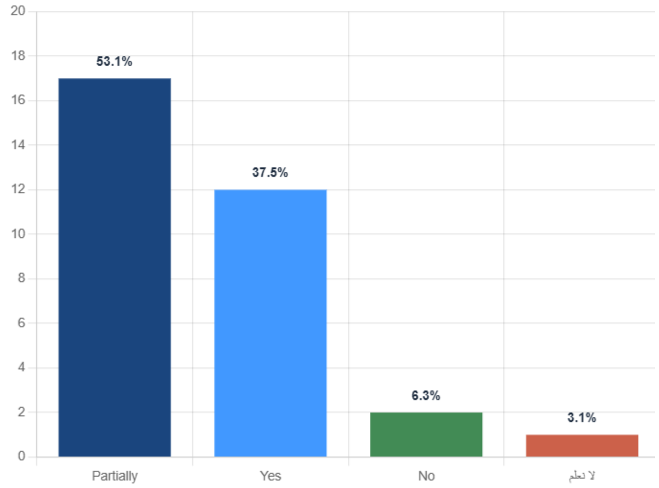


Figure 11 Capacity of current institutional systems for integration with external systems:

Table 5. Data Exchange - key implications for NEDMP implementation Source: Uploaded all-institutions questionnaire response analysis.

Results from questionnaire	Impact on NEDMP	Implementation priority
Official requests are used by 81.3% of respondents.	Data exchange remains procedurally heavy.	Move recurring data flows into structured reporting and portal workflows.
40.0% report standardized machine-readable formats; 33.3% partial standardization.	Interoperability is emerging but incomplete.	Define minimum formats, metadata fields and API requirements.
System integration capacity is partly available.	Platform linkage should be phased.	Start with priority datasets and high-readiness institutions.

6 Quality and Documentation

The quality and documentation responses show that QA/QC general procedures inside institutions, applied documentation standards, the used metadata and reliability practices exist at varying levels across institutions. The charts indicate a mix of implemented, partially implemented and absent practices.

This pattern is critical because metadata, traceability and QA/QC determine whether environmental datasets can be trusted, reused, combined and reported nationally and internationally.

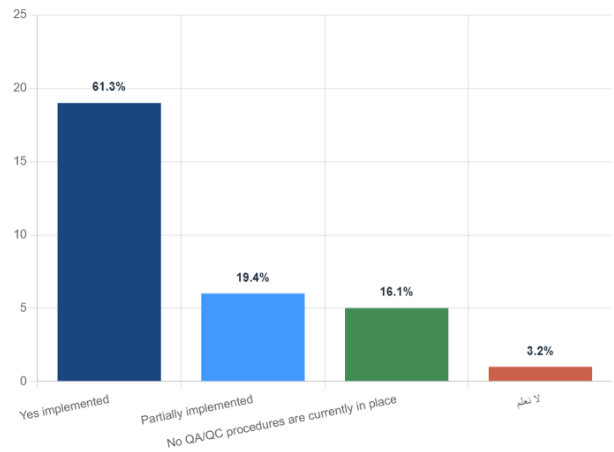


Figure 12 Existence of data Quality Assurance/Quality Control (QA/QC)

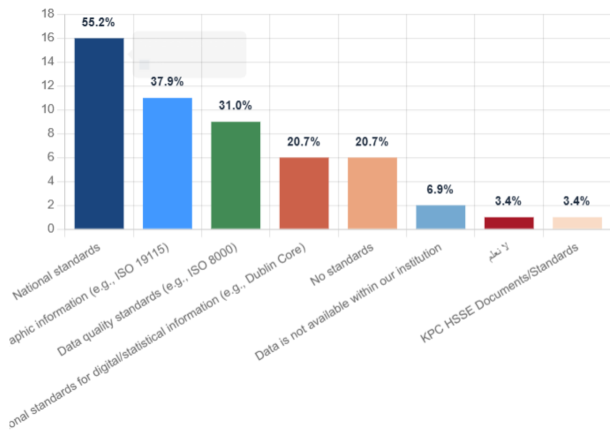


Figure 13 Standards applied for data documentation

The NEDMP should treat quality and documentation as mandatory governance controls. Dataset publication, exchange and national reporting should require minimum metadata, source documentation and QA/QC compliance.

Assessment of the reliability of currently produced or utilized data:

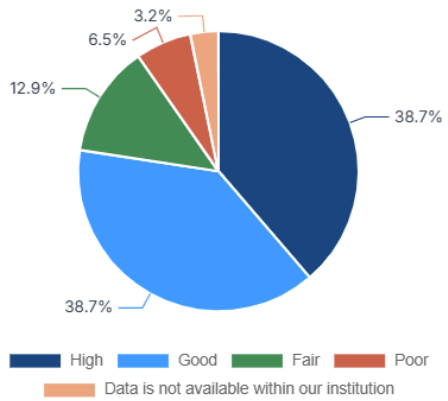


Figure 14 reliability of currently produced or utilized data

Table 4 Quality and Documentation - key implications for NEDMP implementation

Results from questionnaire	Impact on NEDMP	Implementation priority
QA/QC practices are not uniformly implemented.	Data reliability varies across institutions and themes.	Introduce minimum QA/QC rules before data integration.
Metadata and methodology documentation are uneven.	Traceability and auditing may be weak.	Require standard metadata fields for every dataset.
Reliability is assessed in different ways.	Comparability between datasets may be limited.	Adopt a national reliability grading approach.

NEDMP implementation recommendation:

Develop national QA/QC rules, dataset metadata templates, source/method documentation requirements, reliability grading and audit trails for all priority environmental datasets.

7 Systems and Infrastructure

Data related to the systems and infrastructure responses show varied use of environmental data management systems, mixed perceptions of infrastructure efficiency, and uneven backup and disaster-recovery practices.

These findings are important for NEDMP implementation because system integration cannot rely only on environmental policy decisions; it also requires stable updated database, secure backup, data recovery arrangements and institution-level technical readiness.

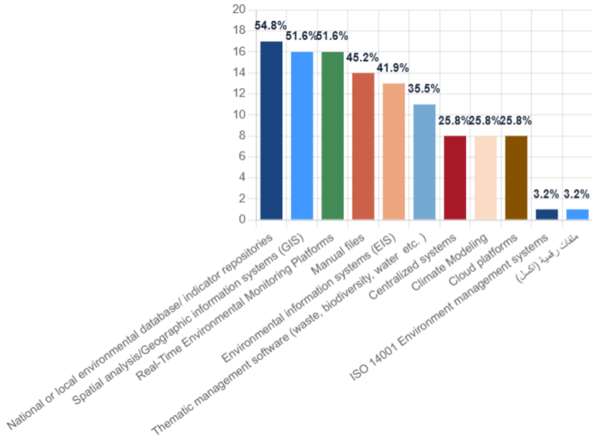


Figure 15 Type of environmental data management systems at the institution

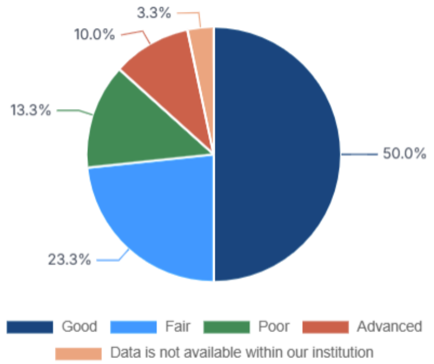


Figure 16 Assessment of institutional infrastructure efficiency for environmental data production and analysis

The NEDMP requires an infrastructure readiness baseline before introducing platform-wide integration. Different institutions may need different pathways: minimum data submission templates, portal-based exchange, or system-to-system linkage.

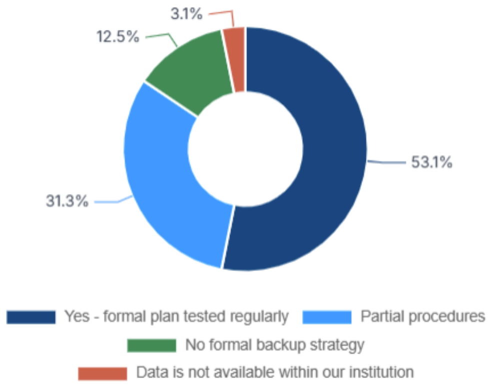


Figure 17 regular data backup and disaster recovery

Results of questionnaire show that institutions currently use different data management systems, which makes integration more complex. The different level of infrastructure efficiency indicates that institutions have different degrees of digital readiness, requiring phased technical support. In addition, the lack of uniform backup and disaster recovery arrangements creates risks for data continuity and sustainability.

NEDMP implementation recommendation

Conduct a national environmental data systems inventory and readiness assessment, covering current systems, data repositories, backup, cybersecurity, integration capacity and hosting arrangements.

Table 5 key implications for NEDMP implementation

Results from questionnaire	Impact on NEDMP	Implementation priority
Institutions use different data management systems.	Integration requires a tiered architecture.	Map systems and repositories before platform design.
Infrastructure efficiency is mixed.	Digital transformation readiness varies by institution.	Use readiness scoring and phased technical support.
Backup and recovery are not uniform.	Data continuity and resilience risks exist.	Set national minimum backup and disaster-recovery requirements.

8 Human Resources

Human resource responses show variation in the number of specialists, staff sufficiency, overall capacity, and training needs. The training needs data indicates demand across environmental data and information systems functions.

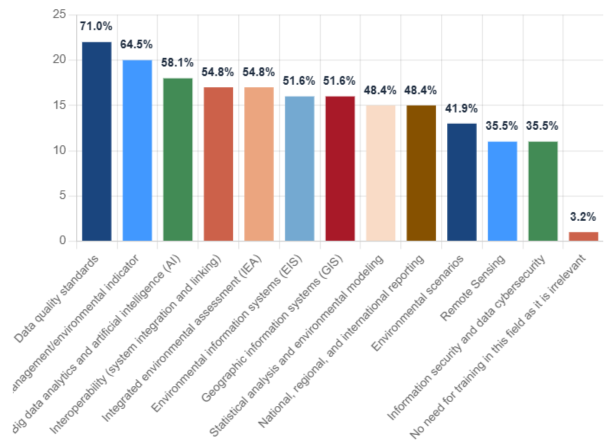


Figure 18 Training needs specific to environmental data and information systems

The human-capital dimension is a central enabling condition. Even where data and systems exist, the NEDMP will not be sustainable without trained data stewards, metadata officers, GIS/data analysts, reporting specialists and system administrators.

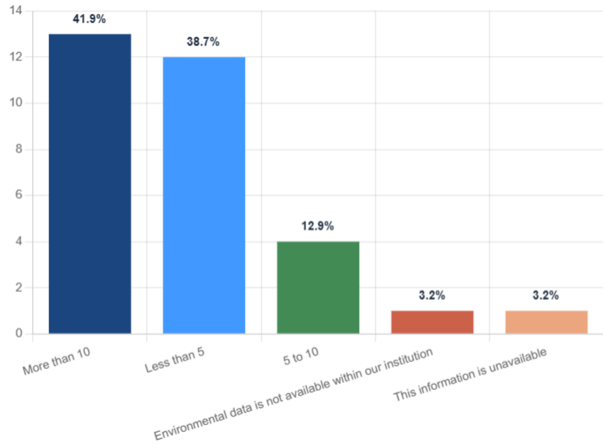


Figure 19 Number of specialists working in the field of environmental data and indicators

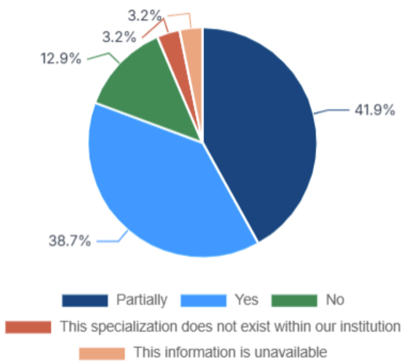


Figure 20 number of staff sufficient to meet data management needs

Capacity development is not a supporting activity; it is a core implementation pillar. The NEDMP should institutionalize continuous training and role-specific competency development

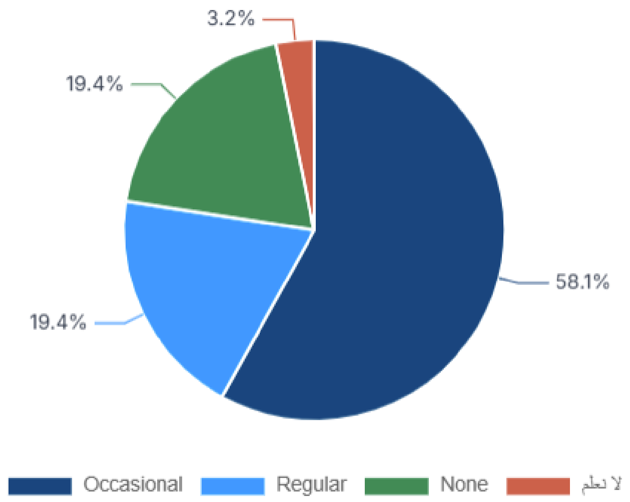


Figure 21 raining programs in environmental data, indicators, and information systems

NEDMP implementation recommendation

Prepare a national capacity-building programme covering data governance, metadata, QA/QC, GIS, databases, APIs, environmental indicators, reporting, dashboard use and digital platform operation.

Table 6 Human Resources - key implications for NEDMP implementation

Results from questionnaire	Impact on NEDMP	Implementation priority
Staff and specialist capacity varies.	Implementation burden may fall unevenly across institutions.	Design a role-based capacity-building pathway.
Training needs are multi-dimensional.	Technical, governance and reporting skills are all needed.	Develop modular training packages and refresher programmes.
Capacity levels influence data quality and sharing.	Human capacity is linked to all NEDMP pillars.	Assign data stewards and train them formally.

9 Governance and Security

Data governance and security include information regarding management policies, clarity of roles, data security systems, decision-making authority, legal reliance for sharing and dissemination, and the existence of documented SOPs.

The pattern of the assessment indicates that governance structures exist but require greater standardization across institutions. Security, legal basis and SOPs should be considered integral to the national environmental data governance framework.

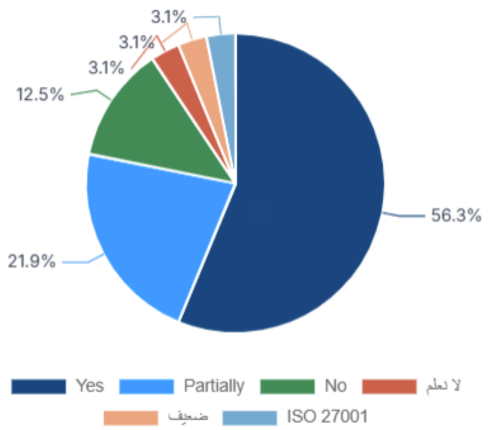


Figure 22 Existence of an Information Management and Data Governance Policy

Clarity of roles and responsibilities regarding data within the Authority:

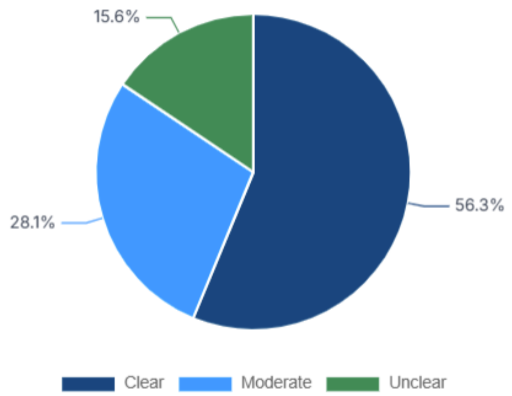


Figure 23 roles and responsibilities regarding data

NEDMP implementation should not depend on informal practices. It requires defined ownership, stewardship, approval authority, legal controls, SOPs and security protocols for environmental data management

NEDMP implementation recommendation

Establish a national environmental data governance framework with role definitions, SOPs, access controls, security protocols, escalation procedures and annual compliance review

Existence of data security systems, protocols, and breach response plans:

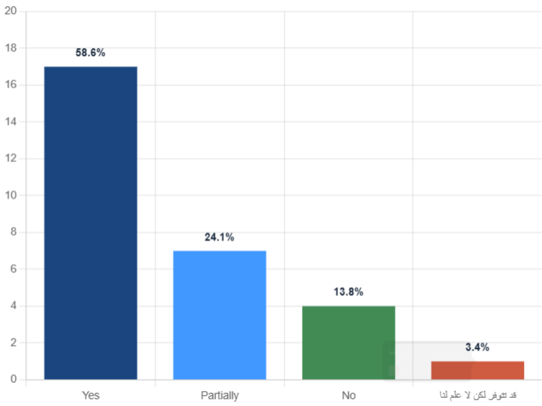


Figure 24 data security systems, protocols, and breach response plans

documented Standard Operating Procedures (SOPs) governing environmental data collection

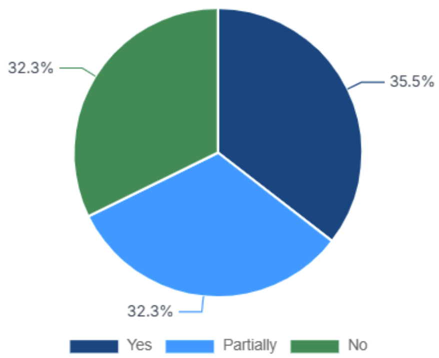


Figure 25 Standard Operating Procedures (SOPs)

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Table 7 Governance and Security - key implications for NEDMP

Results from questionnaire	Impact on NEDMP	Implementation priority
Policies and roles are uneven.	Institutional accountability may be unclear.	Define data owner, steward, custodian and approving authority.
Security and breach-response arrangements vary.	Sensitive data require controlled protection.	Introduce minimum security and access-control rules.
SOP coverage is not universal.	Processes can remain person-dependent.	Develop national SOP templates and institutional adaptation guidance.

10 Reporting and Compliance

Data received with respect to reporting and compliance reflect the participation in environmental reports, mandatory reporting requirements, readiness for reporting, data required from other institutions and compliance/reporting challenges.

The results show that environmental reporting is a major driver of data demand. However, reporting effectiveness depends on data availability, quality, timeliness, inter-institutional coordination and clarity over required indicators.

Reports the institution participates:

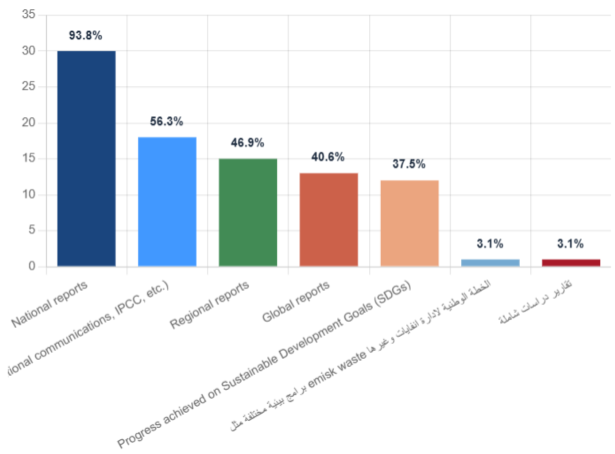


Figure 26 Reports the institution participates

Readiness of institutional data and indicators for Environmental reporting

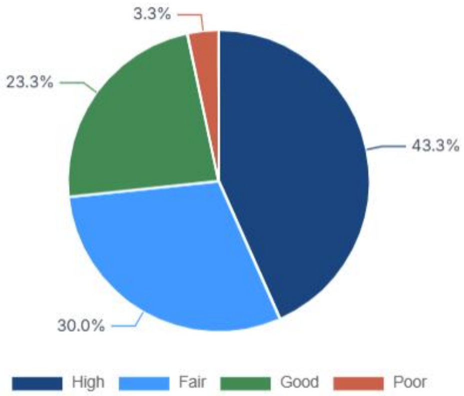


Figure 27 Readiness of institutional data and indicators for environmental reporting

The NEDMP should be aligned with national reporting obligations, including national environmental reporting and other international obligations. Reporting should become a structured data-flow process that contribute to national data collection and benefit from standardize metadata datasets.

Environmental data collected to meet mandatory reporting requirements:

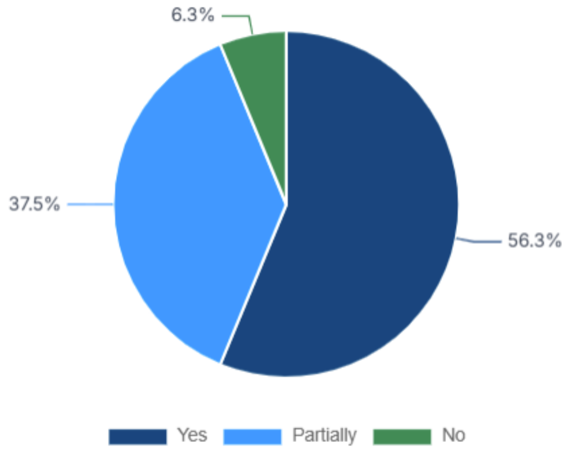


Figure 28 Reporting Obligations

Key challenges

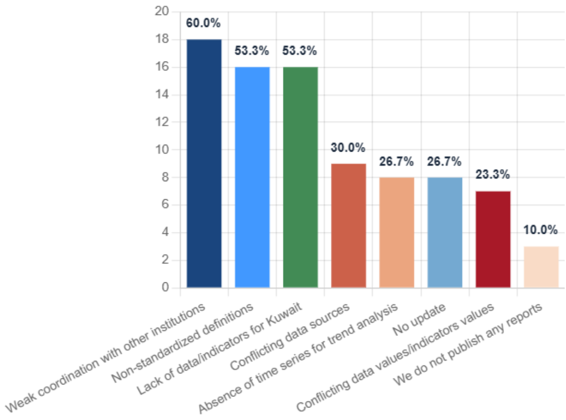


Figure 29 Key challenges for compliance and reporting

NEDMP implementation recommendation

Develop a reporting data requirements matrix that links each report and obligation to datasets, indicators, sources, custodians, update frequency, metadata and QA/QC requirements.

Table 8 Reporting and Compliance - key implications for NEDMP implementation

Results	Impact on NEDMP	Implementation priority
Institutions participate in multiple reporting processes.	Reporting can drive standardization.	Map datasets to reports and indicators.
Data is required from other institutions.	Reporting depends on inter-agency exchange.	Create recurring institutional data provision schedules.
Compliance challenges are visible.	Data gaps can weaken national reporting.	Use NEDMP to close reporting-critical data gaps first.

11 Data Use in Policy

The data-use in policy assessment evaluates whether institutions are using environmental data to support policies and decisions, major areas of use and the stage at which data enters policy development.

Environmental data becomes valuable when it informs decisions. The NEDMP should promote analytical products, indicators and national portals' dashboards that serve policy design, regulatory follow-up and strategic environmental planning.

Environmental data and indicators utilized to support policies and decisions:

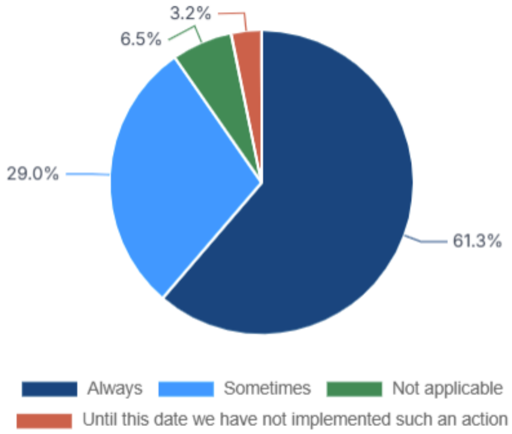


Figure 30 Environmental data and indicators utilized to support policies

The findings highlight the importance of strengthening the pathway from data collection to analysis, interpretation, policy formulation, monitoring and evaluation. The NEDMP should therefore focus not only on data storage, but also on policy relevance and usability.

Major areas for utilizing environmental data and indicators:

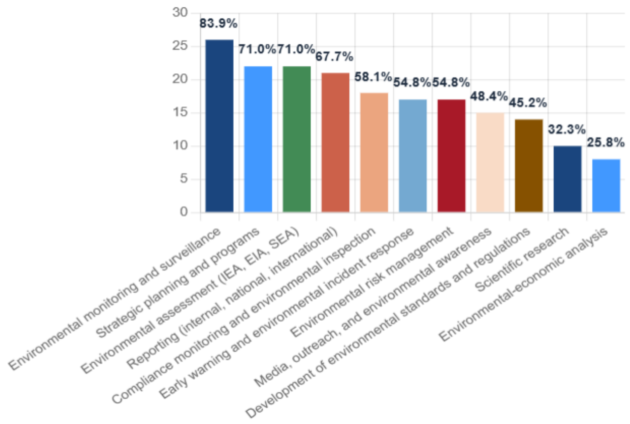


Figure 31 Utilization of environmental data and indicators

Environmental data used in policy development:

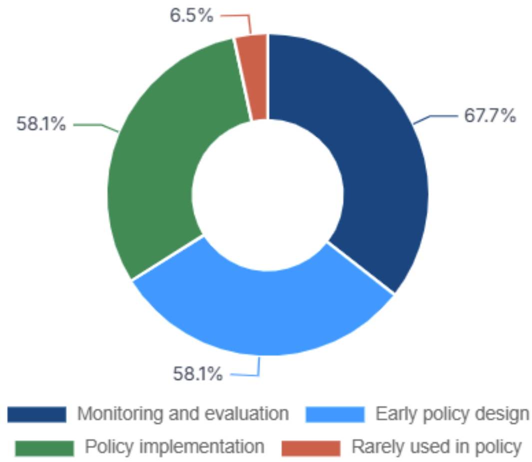


Figure 32 Environmental data used in policy development

The vast majority of institutions (67.7%) use environmental data for monitoring and evaluation, followed closely by early policy design and policy implementation (both 58.1%). Only a small minority (6.5%) report that data is rarely used in policy. This indicates that environmental data is already well-integrated into the policy cycle, providing a strong foundation for the NEDMP to build upon.

NEDMP implementation recommendation

Define priority policy use cases for NEDMP, including environmental reporting, compliance monitoring, pollution control, climate and biodiversity analysis, spatial planning and national development monitoring.

Table 9 Data Use in Policy - key implications for NEDMP implementation

Results	Impact on NEDMP	Implementation priority
Data is used for policies and decisions at different levels.	There is a policy demand for environmental evidence.	Translate datasets into policy-ready indicators and dashboards.
Use areas vary by institution.	The NEDMP should support multiple decision contexts.	Develop use-case based data products.
Policy-stage use is uneven.	Data may not always influence early policy design.	Integrate data requirements into policy cycle and evaluation processes.

12 Digital Platforms

Digital platform responses cover current platform use in the responding institutions, their AI/ML applications, barriers to AI, eMISK utilization, reasons for limited eMISK use, readiness for a unified national platform and reasons for limited readiness.

Utilization of digital platforms for environmental data processing, sharing, or collection:

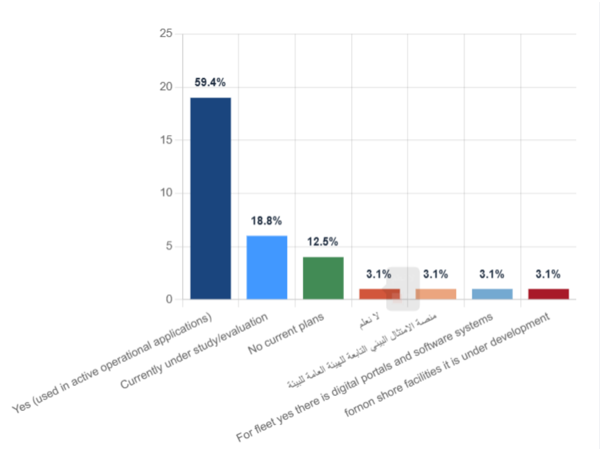


Figure 33 Utilization of digital platforms

The results show that digital transformation is already part of the national discussion, but it must be sequenced carefully. AI and automated integration require high-quality data, metadata, governance rules and secure interoperable systems.

Utilization of Artificial Intelligence (AI) tools or develop machine learning models:

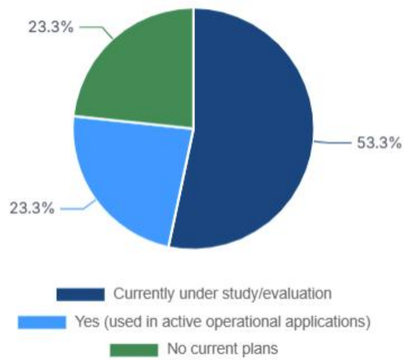
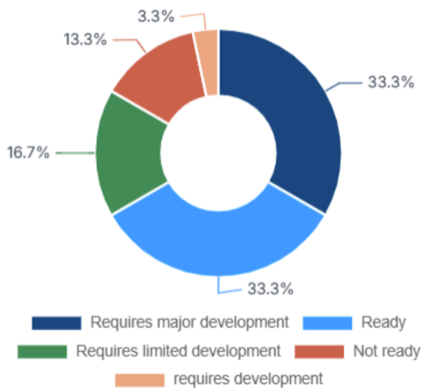


Figure 34 Utilization of Artificial Intelligence (AI)



Major barriers limiting the adoption of AI for environmental purposes:

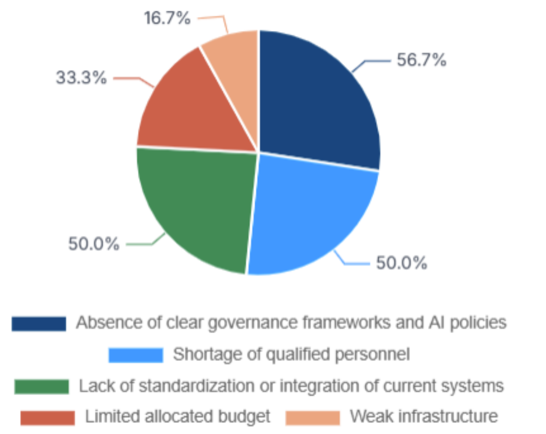


Figure 35 Major barriers limiting the adoption of AI for environmental purposes

Utilization of Environmental Monitoring Information System (eMISK):

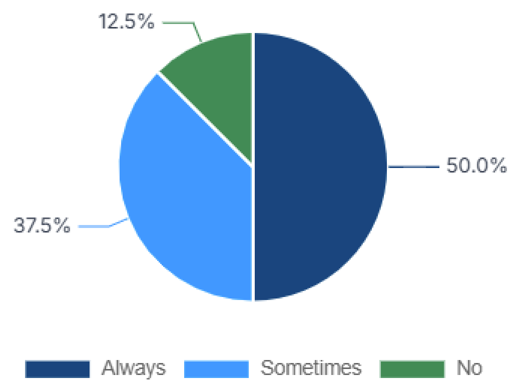


Figure 36 Utilization of Environmental Monitoring Information System (eMISK)

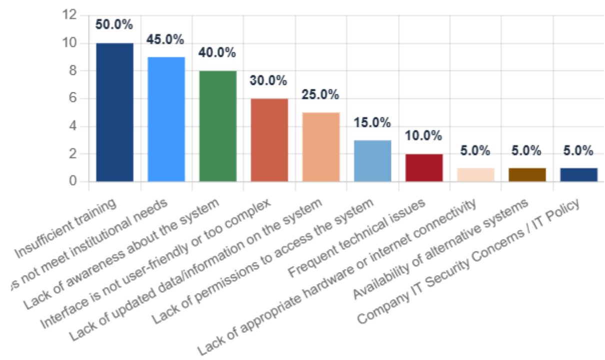


Figure 37 main reasons for not using the (eMISK)

NEDMP implementation recommendation

Develop a digital transformation readiness roadmap for eMISK and the future national environmental data platform, including phased integration, data submission standards, metadata registry and responsible AI principles.

Table 10 Digital Platforms - key implications for NEDMP implementation

Results from questionnaire	Impact on NEDMP	Implementation priority
Digital platform use exists but varies.	A national platform must accommodate different readiness levels.	Use phased integration pathways.
AI adoption faces barriers.	AI cannot precede data quality and governance.	Prepare AI-readiness rules and data maturity thresholds.
eMISK and platform readiness need clarification.	Existing systems should be integrated rather than bypassed.	Assess and support eMISK role and define platform architecture.

13 Funding

Assessment of current Funding responses cover whether resources are allocated for data collection, management and system updates, and the major funding sources for environmental data functions.

Allocated funding for data collection, management, and system updates:

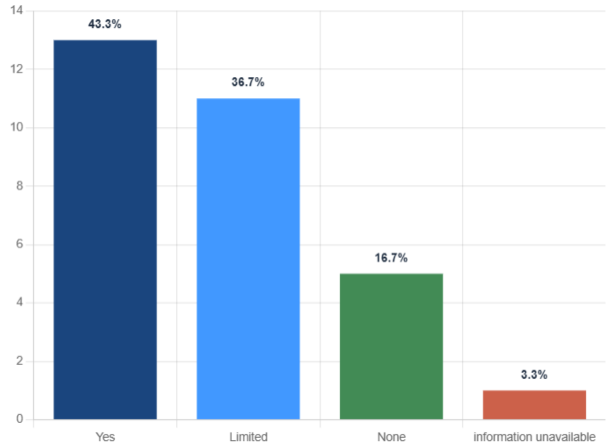
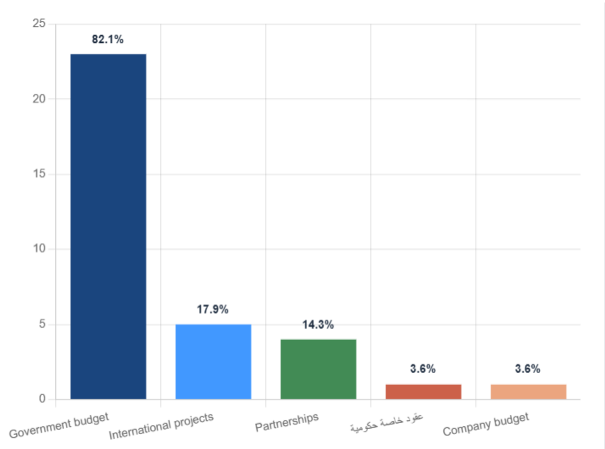


Figure 38 Allocated funding for data collection, management, and system updates

The findings indicate that financing is a core sustainability issue. Data systems require recurring funding, not only one-off project investment. The NEDMP should therefore include budget lines for data governance, systems, maintenance, staff, training and quality control.

Major funding sources for data collection and management:



Without sustainable financing, data quality, platforms, maintenance and capacity-building will remain vulnerable. The NEDMP should include resource mobilization as an implementation pillar.

NEDMP implementation recommendation

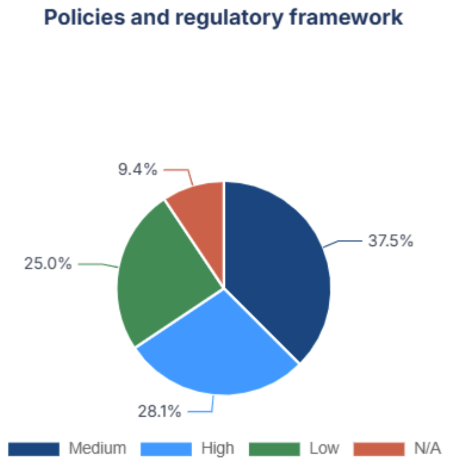
Prepare a costed NEDMP implementation plan with short-, medium- and long-term budget requirements, financing sources, annual operating costs and institutional responsibilities.

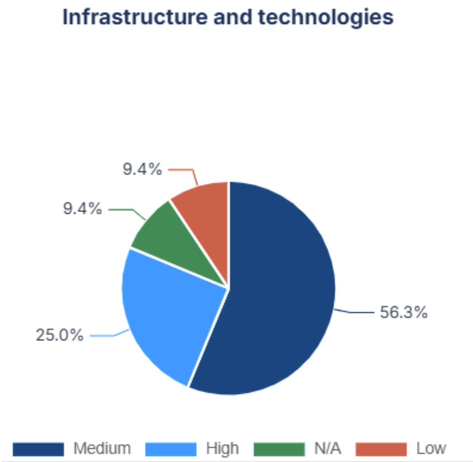
Table 11 Funding - key implications for NEDMP implementation

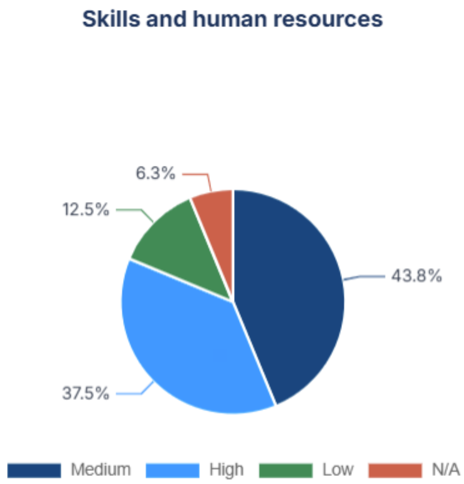
Results	Impact on NEDMP	Implementation priority
Funding allocation varies.	Data management sustainability may be uneven.	Identify minimum annual financing requirements.
Funding sources are not uniform.	Implementation will need coordinated resource mobilization.	Create dedicated budget lines and funding partnerships.
System updates require recurring resources.	Digital transformation cannot rely on capital costs alone.	Include maintenance, training and QA/QC in costed plans.

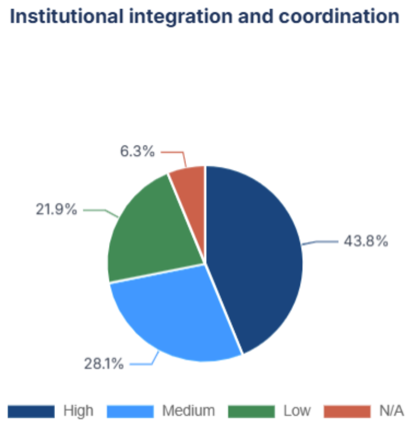
14 Challenges

Identification of current challenges are important to identify the constraints facing policy/regulatory framework, infrastructure and technologies, skills and human resources, institutional integration and coordination, finance and sustainability, and other challenges which are the core barriers to environmental data management and governance.









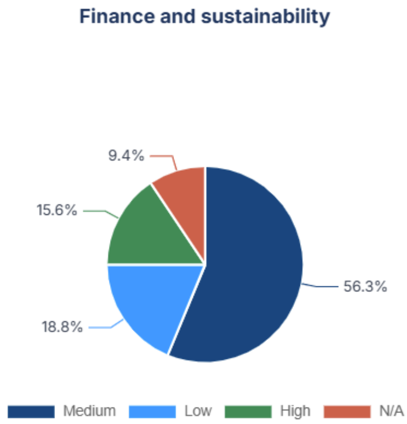
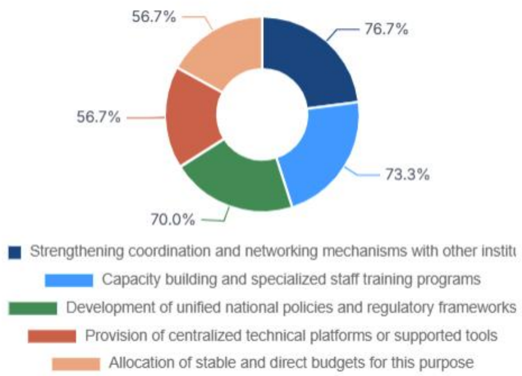


Figure 39 Most significant challenges

Support required to improve environmental data management and governance:



Needs and support data received point to coordination mechanisms, capacity-building, unified policies and regulatory frameworks, centralized technical platforms or supported tools, and direct budgets. Recommendations emphasize standardization, stronger coordination, data-sharing mechanisms and improved digital systems.

The stakeholder needs and recommendations align strongly with the NEDMP pillars: governance, quality, interoperability, sharing/transparency, capacity and sustainability. This confirms that NEDMP implementation should be phased, practical and institutionally owned.

NEDMP implementation recommendation

Translate stakeholder needs into an implementation action plan with responsible entities, deliverables, timelines, KPIs, capacity-building packages and financing requirements.

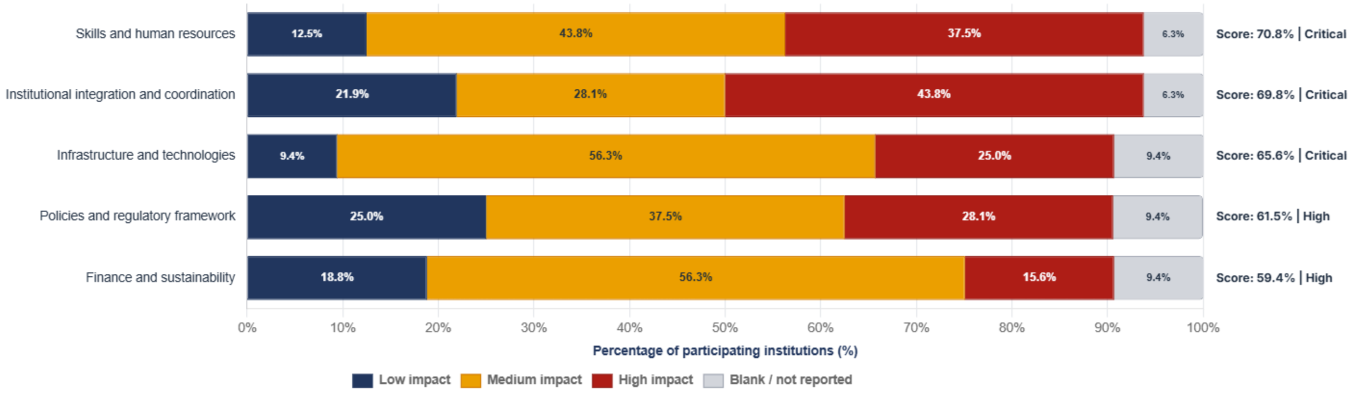


Figure 40 Degree of Challenges Impact

Table 12 Challenges, Needs and Recommendations - key implications for NEDMP implementation

Results from questionnaire	Impact on NEDMP	Implementation priority
Stakeholders identify governance, infrastructure, skills, coordination and financing gaps.	Challenges are institutional and technical, not only data-related.	Implement integrated action packages under the five NEDMP pillars.
Coordination and capacity-building are recurring needs.	Implementation requires a sustained institutional network.	Create a focal-point network and annual capacity programme.
Stakeholder recommendations emphasize standards and systems.	National consistency is a priority.	Adopt common standards, SOPs and platform-readiness rules.

Interlinkage between Existing Challenges and Weaknesses in Consistency, Reliability, and Standardization of Environmental Datasets:

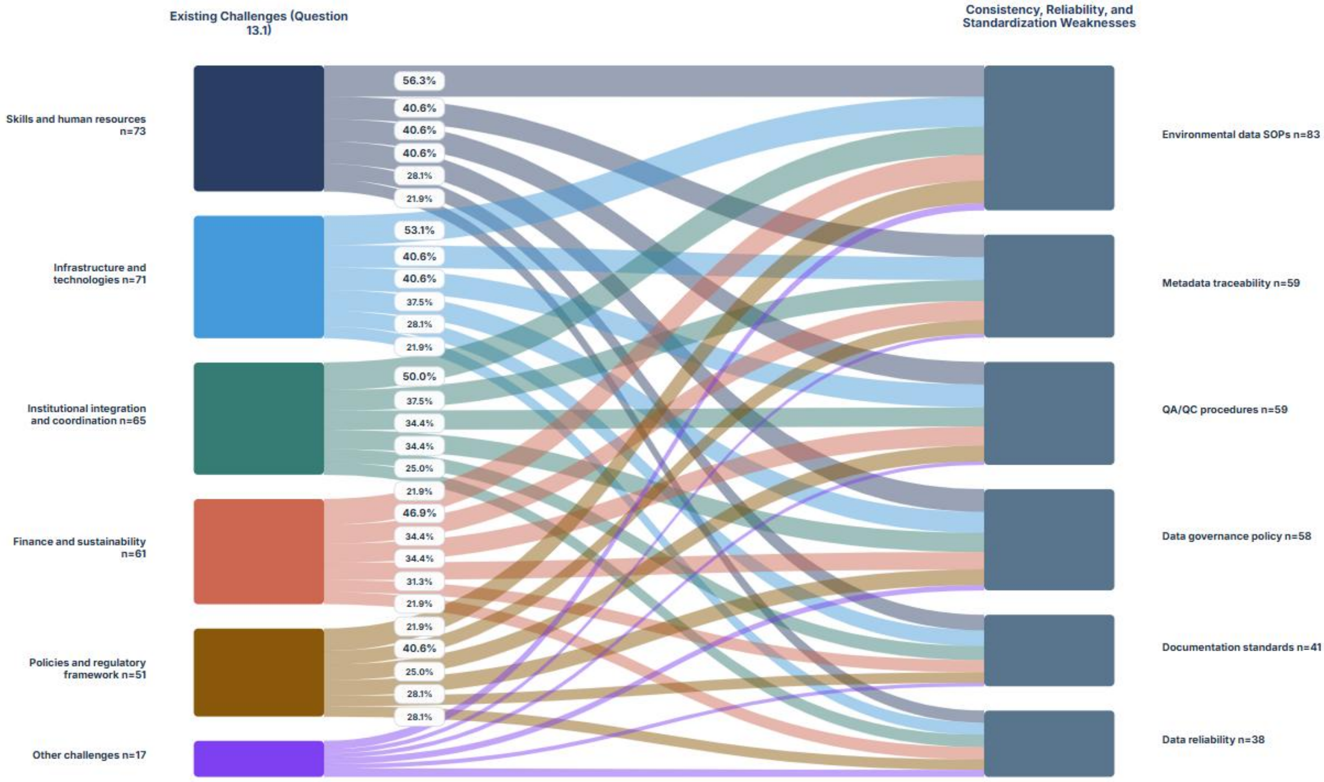


Figure 41 Existing Challenges and Weaknesses in Consistency, Reliability, and Standardization of Environmental Datasets

15 Final Synthesis and Implementation Phases

The data received through the assessment questionnaire responses demonstrate that Kuwait has the institutional participation, data production base and strategic demand required to develop the NEDMP. The most important implementation challenge is to convert dispersed institutional data practices into a coherent national environmental data governance system.

Results show that environmental data production is relatively evident. However, the enabling conditions for national integration are not uniform. Therefore, NEDMP should focus on governance measures, data standards, metadata, QA/QC, access rules, interoperability. The plan must prioritize resolving sustainable financing constraints and human capacity challenges prior to advancing to digital integration or AI-enabled analytics and applications.

The data management plan should be implemented in phases. The short-term phase should establish national data portal to collect and standardize data cross institutions. This initial phase should apply the required procedures for governance and inventory foundations. The medium-term phase should standardize priority datasets and build platform readiness. The long-term phase should support automated data exchange, dashboards, advanced analytics and integrated reporting.

Table 13 consolidated recommendations for NEDMP implementation

Priority area	Recommended action	Expected implementation result
Governance	Create a national environmental data governance committee and institutional focal-point network.	Clear decision-making, accountability and coordination.
Dataset inventory	Prepare a national environmental dataset registry and indicator catalogue.	Known datasets, custodians, sources, update cycles and gaps.
Data quality	Adopt national QA/QC, metadata and reliability standards.	Improved trust, traceability and reuse of environmental data.
Data sharing	Develop access, classification, open-data and controlled-release rules.	Balanced transparency, legal compliance and data protection.
Interoperability	Introduce common data formats, templates, APIs and exchange protocols.	More efficient system-to-system and institution-to-institution exchange.
Digital platform	Define the role of eMISK and the future national environmental data platform.	Phased integration pathway and reduced duplication.
Capacity	Implement role-based training for data stewards, GIS/database teams and reporting officers.	Sustainable institutional capacity for NEDMP operation.
Financing	Prepare a costed implementation plan with recurring budget requirements.	Long-term sustainability for systems, QA/QC, staffing and maintenance.
Monitoring	Adopt NEDMP KPIs and annual progress review.	Measurable implementation and continuous improvement.